

NSDL IT Compliance & Asset Management Portal

A full-stack, agent-less IT compliance and asset management platform built for NSDL branch audits. Automatically discovers, audits, and reports on all Windows workstations across a local network — with zero manual effort on target machines.

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Overview

The **NSDL IT Compliance & Asset Management Portal** is a self-hosted web application that replaces the manual, paper-based compliance audit process at NSDL branch offices.

An IT officer connects their laptop to a branch WiFi network, opens the portal in a browser, and the system:

1. **Discovers** all live devices on the subnet automatically.
2. **Identifies** Windows workstations, mobiles, printers, and unknown devices.

- 3. **Audits** each Windows machine — collecting hardware specs, software inventory, login history, antivirus status, hotfixes, and more.
- 4. **Generates** a structured PDF inspection report and JSON data file.
- 5. **Displays** all collected data in a rich, interactive dashboard.

Tech Stack

Layer	Technology	Purpose
Backend	Python 3 + FastAPI	REST API server, audit ingestion, network scanning
Frontend	Vanilla HTML/CSS/JS	Single-page application dashboard
Report Generation	ReportLab	PDF compliance report generation
Remote Execution	PyPsExec + PyWinRM	Remote notification delivery to target PCs
Network Scanning	Python socket + subprocess	TCP port scan + ICMP ping sweep + ARP discovery
Audit Script	PowerShell (.ps1)	Agentless data collection on target Windows machines
Data Storage	JSON files + PDF	Audit records stored as structured files per device
Connectivity	Windows WLAN API (netsh)	Programmatic WiFi network management

Project Structure

```
NSDL-Compliance-Audit-Portal/
|
├── backend/
|   ├── main.py          # FastAPI application – all API endpoints
|   └── requirements.txt  # Python dependencies
|
├── frontend/
|   └── index.html        # Single-page dashboard (HTML + CSS + JS)
|
├── scripts/
|   ├── audit.ps1        # PowerShell audit collection script (Windows)
|   └── audit.sh          # Shell audit script (Linux/Mac – legacy)
|
├── user_info/           # Audit output storage (JSON + PDF per audit)
├── logs/
|   └── audit_backend.log  # Application & scan logs
|
└── venv/                # Python virtual environment
```

Core Features

1. Device Audit (Compliance Audit Tab)

- **Agentless collection** — no software needs to be installed on target machines.
- The audit PowerShell script is **dynamically served** from the backend with the correct server IP and a unique session `client_id` pre-injected.
- On execution, the script silently collects:
 - **OS Information** — Name, version, architecture, Windows license status.
 - **Hardware Details** — CPU model, total RAM, disk drives and sizes, GPU name/VRAM/driver version, system manufacturer, model, and serial number.
 - **Network Configuration** — IP address, MAC address, subnet, gateway, all NIC adapters.
 - **Security Status** — Antivirus products detected (Windows Defender, third-party AV).
 - **Hotfixes / Windows Updates** — All installed KB patches with dates.
 - **Peripheral Devices** — USB devices, keyboards, mice, scanners, printers, Bluetooth.
 - **Disk Partitions** — All partitions, sizes, types, and boot flags.

- **Compression Utilities** — WinZip, 7-Zip, WinRAR detection.
- **CD/DVD Drives** — Detection of optical drives.
- **Printers** — All connected/networked printers with driver, port, and status.
- **User Accounts** — All local accounts with disabled/enabled status.
- **Software Inventory** — Full list of 100–200+ installed applications with version, publisher, install date, and size.
- **Login History** — Up to 20 most recent interactive logons from the Windows Security Event Log (requires Admin) with logon type and timestamp.
- The collected data is uploaded to the server as a **JSON payload** and saved.
- A **PDF Inspection Report** is automatically generated and saved.

2. **Asset Registry Tab**

- Manually register and track any device with:
 - Device ID / Hostname
 - Asset Tag (e.g., NSDL-AST-2024-001)
 - Owner / Assignee name, Department, Location / Branch
 - Purchase date, price, and warranty expiry
 - Life cycle stage (Active, Maintenance, Retired, Disposed)
 - Vendor and additional notes
- All registered assets are displayed in a sortable, filterable table.
- Supports **Edit** and **Delete** operations.
- **Export to CSV** for offline reporting.

3. **Network Discovery Tab**

- **Automatic subnet detection** from the current WiFi connection.
- **Two-phase discovery engine:**
 - ICMP Ping Sweep** — Pings all 254 IPs in parallel (128 threads) to populate the OS ARP cache. Ensures **all** live devices are found, including mobile phones and firewalled PCs that block TCP ports.
 - TCP Port Scan** — Scans 10 common ports in parallel (64 threads) with configurable timeout.
- **Smart device classification** by open ports and hostname:
 - Windows Host (ports 135 RPC, 445 SMB, 3389 RDP)
 - Web Service / Network Device (ports 80/443)

- Linux / SSH Device (port 22)
- Printer (port 9100)
- Mobile / Firewalled Device (ARP-only, no open ports)
- **ARP fallback** — Discovers any live device with all TCP ports blocked. Broadcast IPs and ff:ff:ff:ff:ff:ff MACs are filtered out.
- **Audit status enrichment** — Cross-references IPs/hostnames against stored audit files to show  Audited or  Unaudited badge.
- **Send Notification button** — Remotely delivers an IT audit notification to any Windows machine using PsExec or WinRM.
- **Copy Manual Command button** — Generates a PowerShell one-liner for manual execution on a target machine.
- Filter by IP, hostname, OS, or username. Export results to CSV.

4. **Device Audits Tab (Hardware + Software + Logins)**

- Dropdown selector populated with all audited devices from `user_info/`.
- On device selection, displays **three rich panels**:
 - Panel 1 — System Specifications:**
 - OS name and build version, CPU, RAM, Storage, Manufacturer/Model, Serial Number, Antivirus, License status badge.
 - Panel 2 — Recent Logins:**
 - Table of last 20 logins: Username, Domain, Logon Type, Timestamp.
 - Logon types: Local Interactive , Cached Interactive , Unlock , Remote (RDP) .
 - Panel 3 — Software Inventory:**
 - Full table of all installed applications.
 - Search/filter by name, publisher, or version.
 - Export to CSV.

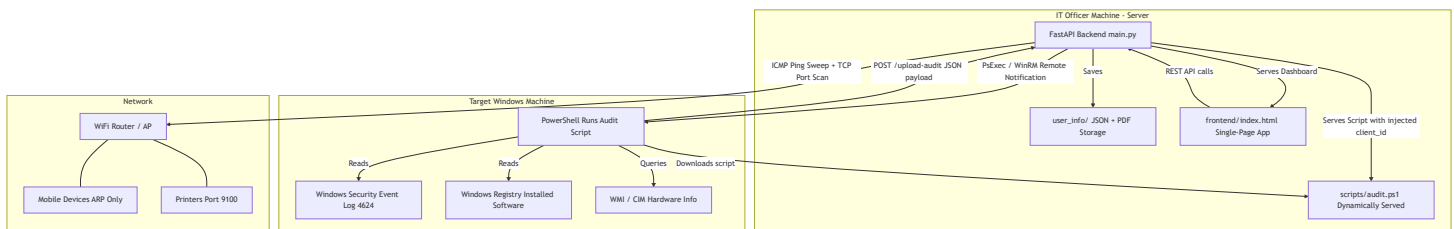
5. **WiFi Dashboard Tab**

- View **available WiFi networks** with SSID, signal strength, and security type.
- **Connect** to any network directly from the browser.
- After connecting, automatically runs a **device scan** on the new subnet.
- Displays live **Connected Devices** table with audit status.

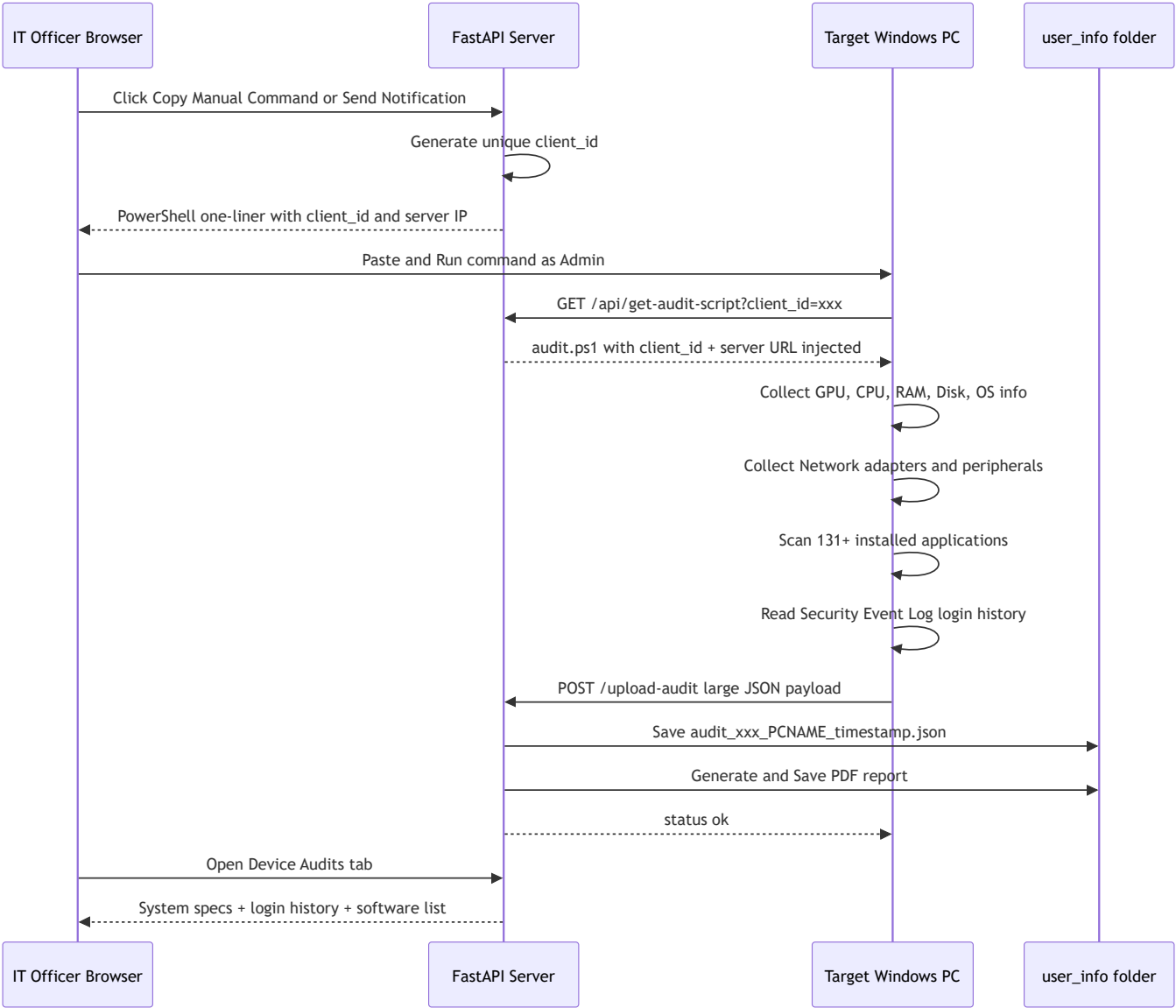
6. PDF Report Generation

- Every audit automatically generates a **professional PDF compliance report** containing:
 - Branch name, code, officer name, date, and audit timestamp.
 - Compliance summary table (OS, CD drive, printer, antivirus, compression).
 - Full hardware specification and network adapter table.
 - Complete hotfix / Windows Update list.
 - Full software inventory table.
- Reports stored as `audit_<id>_<PC name>_<timestamp>.pdf` in `user_info/` .

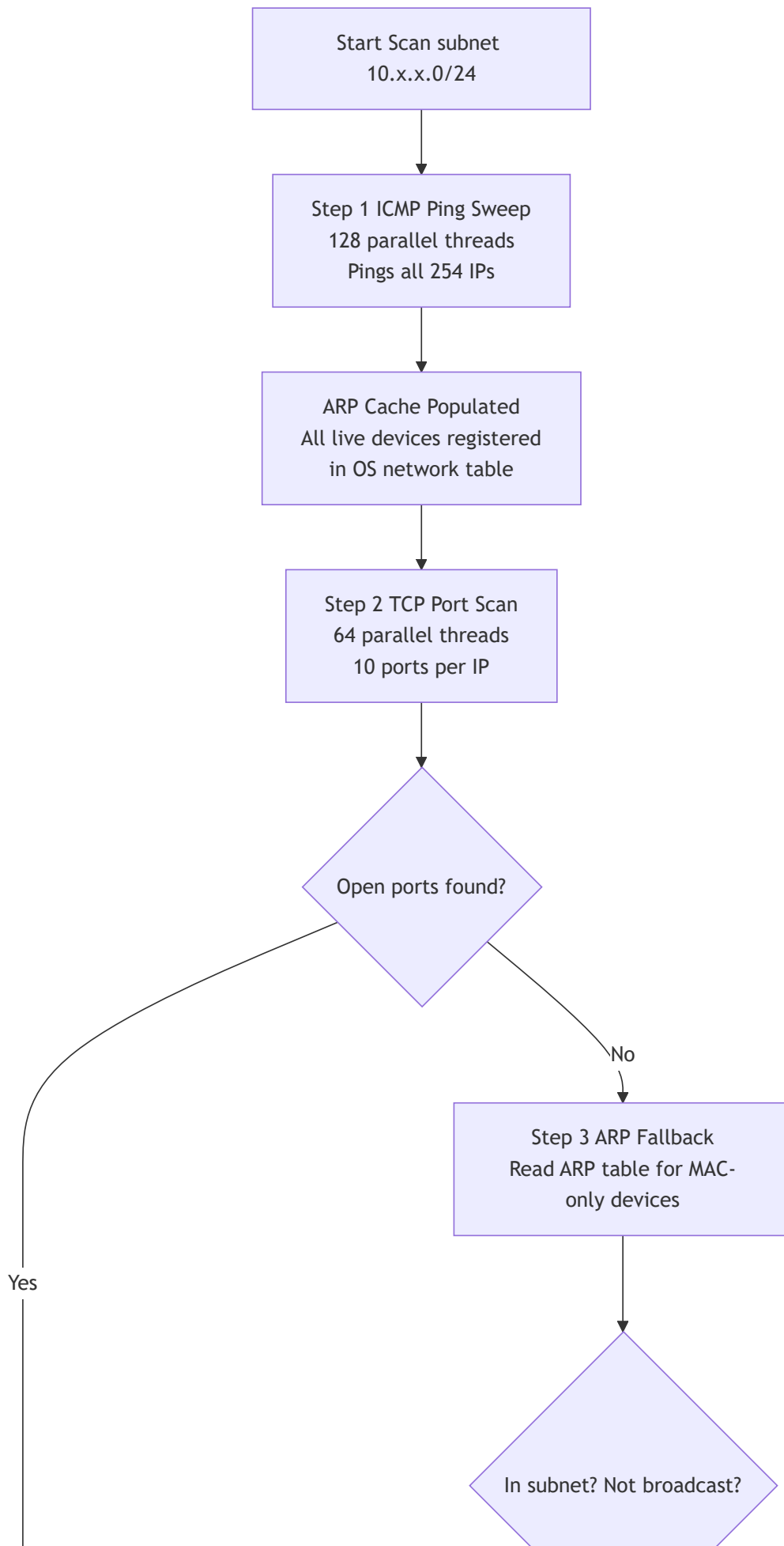
System Architecture

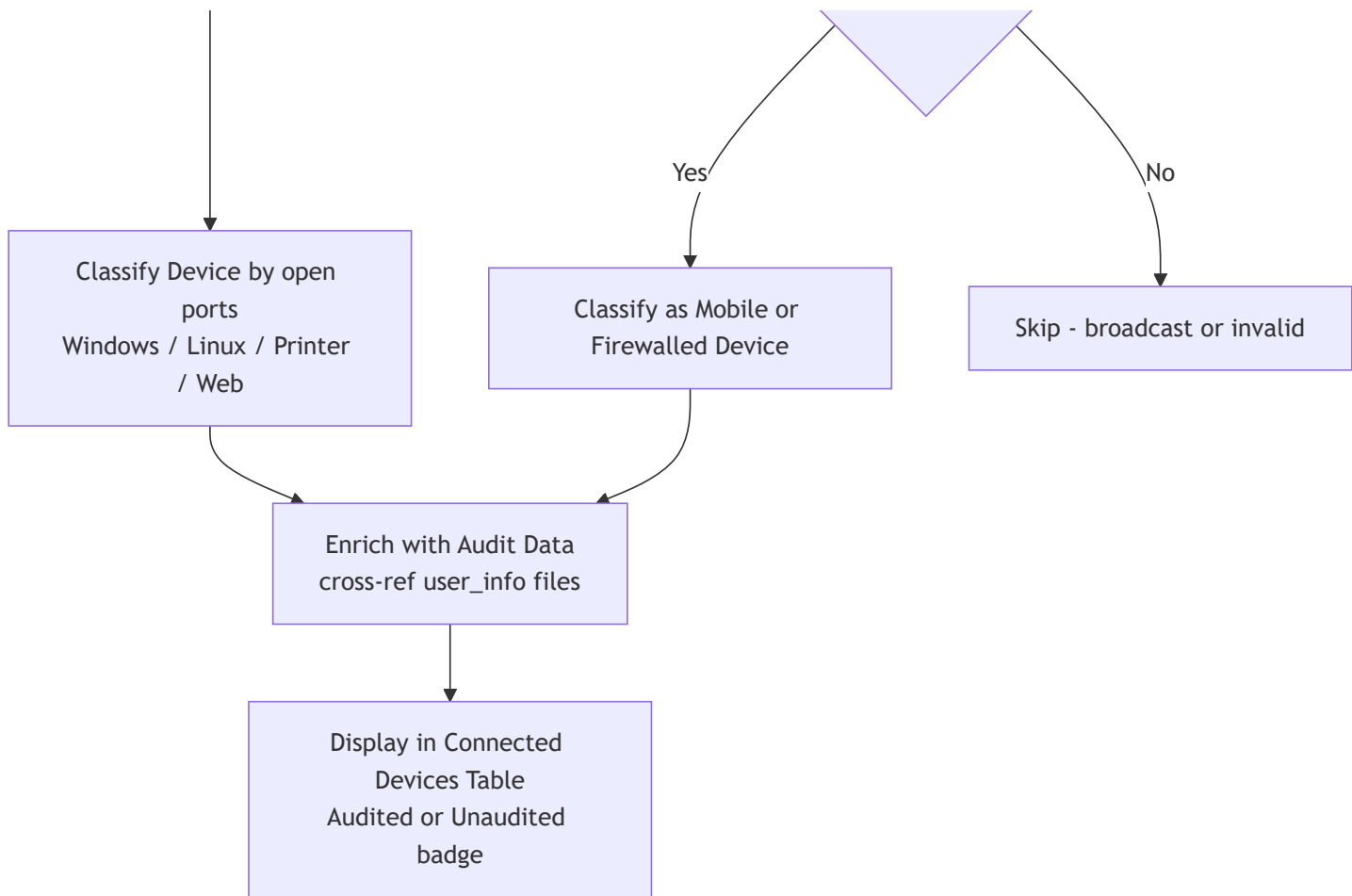


Audit Workflow

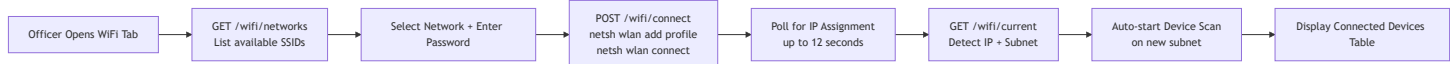


Network Discovery Workflow

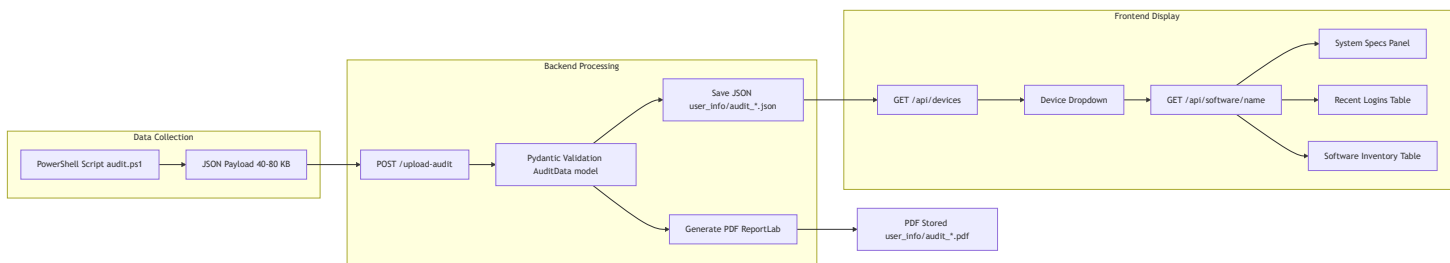




WiFi Management Workflow



Data Flow Diagram



API Reference

Method	Endpoint	Description
GET	/wifi/networks	List all available WiFi networks
GET	/wifi/current	Get current WiFi connection details
POST	/wifi/connect	Connect to a WiFi network
GET	/wifi/scan-devices	Scan current subnet for live devices
POST	/discover/network-scan	Run custom subnet port scan
POST	/upload-audit	Receive and store a full audit payload
GET	/api/get-audit-script	Serve dynamic audit.ps1 with injected client_id
GET	/api/devices	List all audited devices
GET	/api/software/{computer_name}	Get full audit data for a device
POST	/audit/send-notification	Remote-execute audit notification on target PC
GET	/assets	List all registered assets
POST	/assets	Create or update an asset registration
DELETE	/assets/{device_id}	Remove an asset
GET	/scripts/audit.ps1	Serve raw audit script file
GET	/	Serve the frontend dashboard

Setup & Running

Prerequisites

- Python 3.10+
- Windows OS (required for WiFi and network scanning features)
- PowerShell 5.1+ on target audit machines

Installation

```
# 1. Clone the repository
git clone https://github.com/Rudra-Gupta15/Prevoyance_inspection.git
cd Prevoyance_inspection

# 2. Create and activate virtual environment
python -m venv venv
venv\Scripts\activate

# 3. Install Python dependencies
pip install -r backend/requirements.txt

# 4. Start the server
uvicorn backend.main:app --host 0.0.0.0 --reload
```

Accessing the Portal

- **Local machine:** `http://localhost:8000`
- **From any device on the same network:** `http://<your-server-ip>:8000`

Running an Audit on a Target Machine

```
# Run as Administrator on the target PC
powershell -c "Invoke-WebRequest -Uri 'http://<SERVER_IP>:8000/api/get-audit-script?client_id=m"
```

Tip: Use the **Copy Manual Command** button in the WiFi Dashboard tab to auto-generate this command with the correct server IP pre-filled.

Security & Permissions

Feature	Requirement
Run audit script	Any standard user
Read full login history	Administrator on target machine
Send remote notification via PsExec	Admin credentials + SMB/File Sharing enabled

Feature	Requirement
Send remote notification via WinRM	WinRM enabled on target
WiFi connect/disconnect	Sufficient Windows privileges on server
Read Security Event Log	Administrator on target machine

Enabling Remote Audit via PsExec

Run on the **target machine** as Administrator:

```
# Allow remote administration for local accounts
New-ItemProperty -Path "HKLM:\SOFTWARE\Microsoft\Windows\CurrentVersion\Policies\System" `
    -Name "LocalAccountTokenFilterPolicy" -Value 1 -PropertyType DWord -Force

# Enable File and Printer Sharing
netsh advfirewall firewall set rule group="File and Printer Sharing" new enable=Yes
```

Output Files

Every completed audit produces two files in `user_info/` :

File	Format	Contents
audit_<id>_<PC>_<timestamp>.json	JSON	Full raw audit data — hardware, software, logins, network
audit_<id>_<PC>_<timestamp>.pdf	PDF	Formatted compliance inspection report for submission